

Budget Justification

Senior/Key Personnel (Line A)

Salaries for two senior/key personnel are requested. All personnel budgeted on Lines A and B will be employees of the FreeMoCap Foundation, the proposing organization, and will be acting as its employees while performing the responsibilities described below. NSF does not recognize executive titles for salary-rate determination; the rates below therefore correspond to each individual's functional responsibilities, and each is well below the 75th-percentile Bureau of Labor Statistics rate for the corresponding occupation in the geographic area where the work will be carried out. Standard occupational classification codes and live BLS references are listed per person. Person-month calculations use 173.33 hours per month.

The two senior/key personnel are budgeted at different rates because they occupy different occupations, not different ranks. The PI's responsibilities classify under Computer and Information Research Scientists (SOC 15-1221) and the co-PI's under Software Developers (SOC 15-1252). Those occupations carry different BLS wage distributions, and each rate is benchmarked against its own occupation rather than against the other person's.

Jonathan Matthis — Principal Investigator

- **Functional title / SOC:** Computer and Information Research Scientist (SOC 15-1221); BLS OEWS reference: bls.gov/oes/current/oes151221.htm
- **Salary rate:** \$96,375/year; **Effort:** 8.0 calendar months; **Requested:** \$64,250
- **Responsibilities:** Overall scientific and organizational direction of the project. Leads governance scoping (comparison and evaluation of stewardship models), the risk-analysis and security plan, and the core-versus-extension architecture decisions. Serves as Technical Lead for the mandatory I-Corps for PESOSE training, with decision-making authority over project direction, including approximately one day per week of ecosystem-discovery interviewing across the training window. The remaining 4 months of his effort continue on unfunded Foundation activities.

Aaron Cherian — co-Principal Investigator

- **Functional title / SOC:** Software Developer (SOC 15-1252); BLS OEWS reference: bls.gov/oes/current/oes151252.htm
- **Salary rate:** \$85,667/year; **Effort:** 9.0 calendar months; **Requested:** \$64,250
- **Responsibilities:** Leads the technical scoping work: definition of the community-extension architecture, design of the validation, benchmarking, and regression-testing infrastructure, and coordination of the data-quality and provenance metadata work. Serves as Entrepreneurial Lead for the I-Corps for PESOSE training, coordinating stakeholder outreach, discovery interviews, and weekly reporting, at approximately one day per week across the training window. The remaining 3 months of his effort continue on unfunded Foundation activities.

Salary support in excess of two months

The PESOSE program does not enforce the general two-month limit on senior personnel salary support, but requests exceeding it require justification under PAPPG II.D.2.f(i)(a). The Foundation defines “year” as the 12-month project period of this award, and applies that definition consistently across all personnel and all NSF proposals.

The requested effort exceeds two months because the employment situation differs from the academic case the policy addresses. NSF's premise is that research is a normal function of faculty and that compensation for it is already included in an institutional salary. Neither individual holds

an academic appointment or any other salaried position from which this effort could be drawn, and this award is the sole source of compensation for the work described. The requested salary is not supplemental to an existing salary; it is the compensation for the work itself.

The scope of effort is also set by the program. PESOSE requires a minimum of 100 ecosystem-discovery interviews and approximately one day per week of interviewing from both the Technical Lead and the Entrepreneurial Lead, plus kick-off, weekly sessions, and a closing meeting. The PESOSE budget guidance directs proposers to budget salary support sufficient to accommodate it.

Other Personnel (Line B)

Paul Matthis — Research Infrastructure Engineer

- **Functional title / SOC:** Network and Computer Systems Administrator (SOC 15-1244); BLS OEWS reference: bls.gov/oes/current/oes151244.htm
- **Salary rate:** \$59,000/year; **Effort:** 9.0 calendar months; **Requested:** \$44,250
- **Responsibilities:** Builds and maintains the technical infrastructure on which the planning activities depend: release and continuous-integration infrastructure (GitHub Actions), project web services, and the Google Cloud Platform environment hosting project telemetry, including the provenance and data-quality instrumentation scoped in the Risk Analysis and Security Plan. Implements the technical environment for the community education pilots, and authors and maintains the contributor-facing operational documentation for that infrastructure. This is a technical position; no administrative or clerical effort is direct-charged to this award.

Fringe Benefits (Line C)

Fringe benefits are requested at a rate of 20% applied to a base of \$172,750, comprising all salaries requested on Lines A and B. Requested fringe: \$34,550. The rate is composed as follows:

Component	Rate	Basis
Statutory payroll taxes (FICA: Social Security and Medicare)	7.65%	Federal statutory rate
Employer contribution toward individual health insurance coverage	9.35%	Estimated employer contribution
Employer retirement plan contribution	3.00%	Estimated employer contribution
Total fringe rate	20.00%	

Beyond the statutory payroll tax component, the rates above are estimates. The Foundation is standing up employer-of-record, payroll, and benefits administration ahead of the period of performance, and the health and retirement contribution levels will be set in that work; that administrative capacity is funded through indirect cost recovery rather than charged directly to this award. The 20% composite rate applies uniformly to all project personnel. The award covers a single 12-month period; no salary escalation is applied or requested.

Equipment (Line D)

No equipment is requested. Funding requests for equipment are not permitted for PESOSE Track 1 proposals.

Travel (Line E)

Total travel requested is \$10,000, all domestic: \$7,000 for two conference trips supporting community building and dissemination, and \$3,000 for I-Corps ecosystem-discovery travel. No

international travel is requested; the program does not permit it. Costs are budgeted under 2 CFR § 200.475.

Trip 1 — American Society of Biomechanics (ASB) annual meeting, 1 traveler (PI). Engagement with the biomechanics research community that constitutes FreeMoCap’s core academic user base; dissemination of results and recruitment of contributors and ecosystem partners.

Trip 2 — Society for Neuroscience (SfN) annual meeting, 1 traveler (co-PI). Engagement with the neuroscience research community, a major user population for accessible motion measurement, and a venue for ecosystem-discovery interviews with prospective users and contributors.

Cost element	Trip 1 (ASB)	Trip 2 (SfN)	Total
Airfare (domestic, round trip)	\$700	\$750	\$1,450
Lodging (4 nights)	\$1,100	\$1,200	\$2,300
Meals and incidentals (5 days)	\$425	\$425	\$850
Conference registration	\$900	\$750	\$1,650
Ground transportation	\$375	\$375	\$750
Trip total	\$3,500	\$3,500	\$7,000

No booth, exhibiting, or other marketing-related expenses are requested.

Ecosystem-discovery travel (I-Corps for PESOSE) — \$3,000. Four domestic trips at approximately \$750 each (airfare or ground transportation, one night lodging, meals and incidentals) to conduct in-person interviews during the training window. The training requires a minimum of 100 discovery interviews, and NSF’s I-Corps guidance expressly permits domestic travel to conduct them in person.

Participant Support (Line F)

No participant support costs are requested. The education and onboarding pilots are conducted with partner organizations that host their own learners; no stipends, subsistence, travel allowances, or registration fees are paid to or on behalf of pilot participants. I-Corps discovery interviewees are likewise uncompensated, consistent with the program’s prohibition on incentives for interviewees.

Other Direct Costs (Line G)

Materials and Supplies — \$8,000

Multi-camera capture kits (consumer-grade webcams and mounting hardware) and workshop materials for the hands-on education pilots with partner organizations, and for hardware-configuration testing across supported platforms. All are consumable or low-cost items; no single item approaches the \$5,000 per-unit federal threshold for equipment. This request is under 10% of the total budget.

I-Corps for PESOSE — \$8,500 total across budget lines

Participation in the mandatory I-Corps for PESOSE training is budgeted at \$8,500, within the \$30,000 program ceiling. Its components fall on three different budget lines and are gathered here for review:

- I-Corps participation fee: \$2,500 — this line (Other Direct Costs)
- Industry Mentor stipend: \$3,000 — Consultant Services, below
- Ecosystem-discovery travel within the training window: \$3,000 — Travel (Line E), above

Salary support for the Technical and Entrepreneurial Leads' I-Corps effort is carried on Line A and excluded from the \$8,500 above to avoid double-counting. The training team is the Technical Lead (Matthis), Entrepreneurial Lead (Cherian), and Industry Mentor (Dahen) — three individuals, meeting the requirement that Track 1 teams budget for at least three and no more than five. No gifts, meals, marketing expenses, or survey tools are requested.

Consultant Services — \$32,000

Track 1 is a scoping and planning award, and several of the mandated planning areas call for independent professional expertise the project team does not itself hold. Consultants are engaged to produce assessments, evaluations, and plans — not to develop the open-source product, which is outside the scope of a Track 1 award. None is an officer or employee of the proposing organization. Expertise, primary organizational affiliation, normal daily compensation rate, and number of days of expected service are given for each below, per PAPPG II.D.2.g(v)(c). No consultant will receive more than \$50,000 over the award period. The Foundation is a nonprofit organization; no owners or equity holders exist whose compensation would be restricted.

Consultant / expertise	Daily rate	Days	Total
Open-source governance and licensing counsel	\$1,800	10.0	\$18,000
Independent security and supply-chain assessment	\$2,000	4.5	\$9,000
Community documentation and education-pilot facilitation	\$500	4.0	\$2,000
I-Corps Industry Mentor (Endurance Idehen)	\$500	6.0	\$3,000
Total		24.5	\$32,000

Legal services are budgeted here rather than as a separate line because NSF treats professional service costs and consultant services as the same category. The legal work requested is specific to the project's scope of work — the licensing and governance analysis the solicitation requires Track 1 proposals to produce. General organizational legal costs, including incorporation, nonprofit compliance, board governance, and employment law, are treated as general administration, are recovered through the indirect cost rate, and are not direct-charged to this award.

Open-source governance and licensing counsel — \$18,000. Independent legal and policy expertise supporting the Organization and Governance planning area, which the solicitation requires Track 1 proposals to address: comparative evaluation of stewardship and fiscal-host models, licensing options for the core product and community-contributed extensions, trademark and project-naming policy, and contributor licensing terms. Deliverable is a written governance and licensing options analysis feeding the ecosystem's organizational design.

Independent security and supply-chain assessment — \$9,000. External security expertise supporting the Risk Analysis and Security Plan, which the solicitation requires Track 1 proposals to address. Scope: threat modeling, review of the dependency graph and software-bill-of-materials posture, evaluation of secure release and chain-of-custody practices against Open Source Security Foundation and CISA/NSA supply chain guidance, and identity and access management review. Deliverable is a written vulnerability and risk assessment with prioritized remediation planning, produced independently of the development team.

Community documentation and education-pilot facilitation — \$2,000. Compensated engagement of experienced community members, budgeted across several individuals, for work that the Community Building planning area depends on: authoring and reviewing onboarding and contributor documentation, facilitating the education and onboarding pilots with partner organizations, and structured evaluation of the contributor experience. This funds documentation,

facilitation, and assessment — not product feature development. The balance of the work is carried by the Research Infrastructure Engineer on Line B, whose effort was increased for that purpose.

I-Corps Industry Mentor — \$3,000. Endurance Dahren, an experienced open-source ecosystem leader, serves as Industry Mentor on the I-Corps for PESOSE training team. NSF's I-Corps guidance requires a few hours per week across the training window in addition to program meetings; the request reflects 6.0 days at \$500/day. He is not an employee of the Foundation and is therefore budgeted here rather than on Line A or B.

Other — \$2,500

The I-Corps for PESOSE participation fee, described above.

Employer-of-record, payroll, and benefits administration services are not requested as direct costs. Those are general administrative functions of the organization, recovered through the indirect cost rate below, and consistent treatment under 2 CFR § 200.414(f) prohibits charging them both directly and through the de minimis rate.

No subawards are requested.

Indirect Costs (Line I)

The FreeMoCap Foundation does not have a federally negotiated indirect cost rate agreement and elects the de minimis rate of 15% of modified total direct costs, as permitted by the PESOSE budget guidance and 2 CFR § 200.414(f).

Modified total direct costs are \$259,800, equal to total direct costs. No equipment, participant support, tuition remission, or subawards are requested, so none of the MTDC exclusions in 2 CFR § 200.1 apply to this budget. Indirect costs: \$38,970.

Budget Summary

Line	Amount
A. Senior/Key Personnel (2 people, 17.0 cal months)	\$128,500
B. Other Personnel (1 person, 9.0 cal months)	\$44,250
C. Fringe Benefits (20% of \$172,750)	\$34,550
D. Equipment	\$0
E. Travel (domestic; international \$0)	\$10,000
F. Participant Support	\$0
G.1 Materials and Supplies	\$8,000
G.3 Consultant Services	\$32,000
G.6 Other (I-Corps participation fee)	\$2,500
G. Total Other Direct Costs	\$42,500
H. Total Direct Costs	\$259,800
I. Indirect Costs (15% of \$259,800 MTDC)	\$38,970
J. Total Requested	\$298,770

The total request of \$298,770 is within the \$300,000 Track 1 ceiling for a 12-month period of performance.